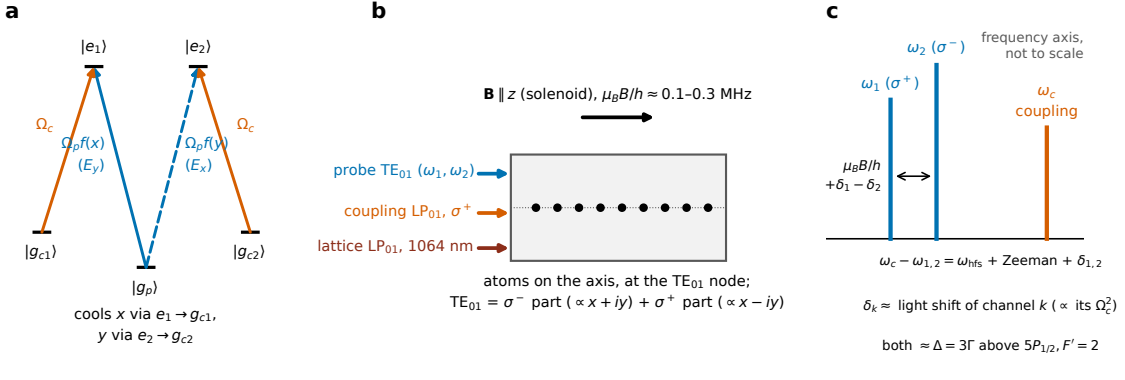
a
$|e_1\rangle$
$|e_2\rangle$
Ω_c
$\Omega_p f(x)$
(E_y)
$\Omega_p f(y)$
(E_x)
Ω_c
$|g_{c1}\rangle$
$|g_{c2}\rangle$
$|g_p\rangle$
cools x via $e_1 \to g_{c1}$,
y via $e_2 \to g_{c2}$
b
$\mathbf{B} \parallel z$ (solenoid), $\mu_B B/h \approx 0.1$–0.3 MHz
probe TE_{01} (ω_1, ω_2)
coupling LP_{01}, σ^+
lattice LP_{01}, 1064 nm
atoms on the axis, at the TE_{01} node;
$TE_{01} = \sigma^-$ part ($\propto x + iy$) + σ^+ part ($\propto x - iy$)
c
frequency axis,
not to scale
ω_2 (σ^-)
ω_1 (σ^+)
ω_c
coupling
$\mu_B B/h$
$+\delta_1 - \delta_2$
$\omega_c - \omega_{1,2} = \omega_{\mathrm{hfs}}$ + Zeeman + $\delta_{1,2}$
$\delta_k \approx$ light shift of channel k ($\propto$ its Ω_c^2)
both $\approx \Delta = 3\Gamma$ above $5P_{1/2}, F' = 2$